UNIVERSITY OF SOUTHEASTERN PHILIPPINES

AR Motion Smasher

A Computer Vision-Based Gesture Recognition System for Mobile Augmented Reality Gaming

A PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR

ICE 322 Professional Elective 2

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III. Introduction

As mobile gaming continues to evolve at an unprecedented pace, it seems that the majority of mobile games have embraced the touch-based model of control. However, as mobile technology matures, game designers have new opportunities to exploit more complex interactions and immersive experiences by physically engaging gamers through the augmented reality (AR) technology. AR Motion Smasher, a proposed AR mobile game for the Android platform will utilize Java and Android Studio for development.

Instead of using traditional control methods that are strictly passive in design and execution, this game will leverage an innovative hybrid software model comprising a separate embedded component implemented written within Python using OpenCV libraries that will process live video feeds coming from the front-facing camera on the device. This embedded component will process the video from the live feed and identify the position of a player's hand or brightly colored marker in the video image, returning the player's continuous position in the form of x/y coordinates. These coordinates will serve as the basis for the collision detection and core gameplay logic in the Android game engine that will enable players to hit virtual objects that are falling.

This work will serve to combine multiple disciplines of advanced mobile application development, human-computer interaction (HCI), and the innovative field of exergaming, which is focused on preventing a sedentary lifestyle by incorporating physical activity into play.

IV. Problem Statement

- **Identified Real-World Problem:** The main type of mobile gaming takes place through passive interaction (touch screen device) thus promoting a sedentary lifestyle. In addition, there are significant barriers to development associated with building natively complex image processing tasks directly into a Java-based Android application as it has a large computational cost.
- **Importance:** Addressing this problem is vital because lack of physical activity is linked to negative health consequences such as obesity and musculoskeletal problems (Cureus, 2025; Academicmed, 2023).
- **Affected Populations:** Children, students, and casual mobile gamers.

V. Project Objectives

General Objective

- To develop an engaging AR mobile game that successfully integrates a Java-based game engine with a Python/OpenCV hand-tracking module to promote physical activity and replace touchscreen controls with physical body movements.

Specific Objectives

- To design an interactive user interface featuring a live camera feed background.
- To implement game mechanics, visual feedback, and a scoring system based on physical interactions.
- To build an OpenCV pipeline using HSV thresholding to track a player's hand or colored marker in real-time.
- To establish a low-latency communication link between the Python image processing module and the Java frontend using the Chaquopy SDK or a local socket server.

VI. Scope and Limitations

Scope

- **Platform:** Android devices running Android 8.0 (Oreo) or higher.
- **Target Users:** Students and casual gamers.
- **Features:** Live AR background tracking a single object (hand or marker), 2D sprite rendering, collision detection, and score tracking.

Limitations

- Designed exclusively for indoor environments with stable lighting (poor lighting degrades computer vision accuracy).
- Limited to offline single-player mode only (no multiplayer).
- Relies on basic 2D graphics overlaid on the camera feed (no 3D assets).

VII. Proposed Solution (Game Concept)

- Proposed Solution: With AR Motion Smasher players are required to actively move their arms while playing therefore will also limit being in a standard sedentary position when mobile gaming.
- Uniqueness: The screen that you see through your camera is populated with virtual objects that fall from the sky onto your screen. The python code will use the BGR frames being captured from the camera to find the center point of your hand and then send those coordinates to Java. This project is unique due to the fact that it uses a dual-language hybrid design; the OpenCV computations are processed by python while the rendering of objects on the screen (i.e. the graphics / output) is handled by Java.

VIII. Target Users

Children, students, and casual adult gamers looking for an interactive and active digital experience.

IX. Game Description

- **Game Genre:** Augmented Reality (AR) Arcade / Action.
- **Game Mechanics:** Virtual targets (meteors, shapes) spawn at the top of the screen and fall downwards over the live camera feed. The system tracks the centroid of the player's hand. If the X/Y coordinates of the hand intersect with the falling target's hitbox, the target is destroyed.
- **Rules of the Game:** The player must intercept the objects before they hit the bottom of the screen. Missing an object deducts one life.
- **Levels or Stages:** The game features progressive difficulty. As the player's score increases, the falling objects move faster and spawn more frequently.
- **Scoring System:** 10 points are awarded per destroyed target.

X. Features of the Game

- Live front-facing camera integration.
- Cross-language real-time image processing.
- Multiple levels of increasing difficulty.
- Score and life tracking system.
- Visual and audio feedback upon collision.

XI. Tools and Technologies

- **Programming Languages:** Java (Game Frontend) and Python 3.10+ (Image Processing Backend).
- **Development Platform:** Android Studio.
- **Libraries/Tools:** OpenCV (cv2), NumPy, Chaquopy SDK (Python-in-Android plugin), and Android CameraX API.

XII. System Flow / Game Flow

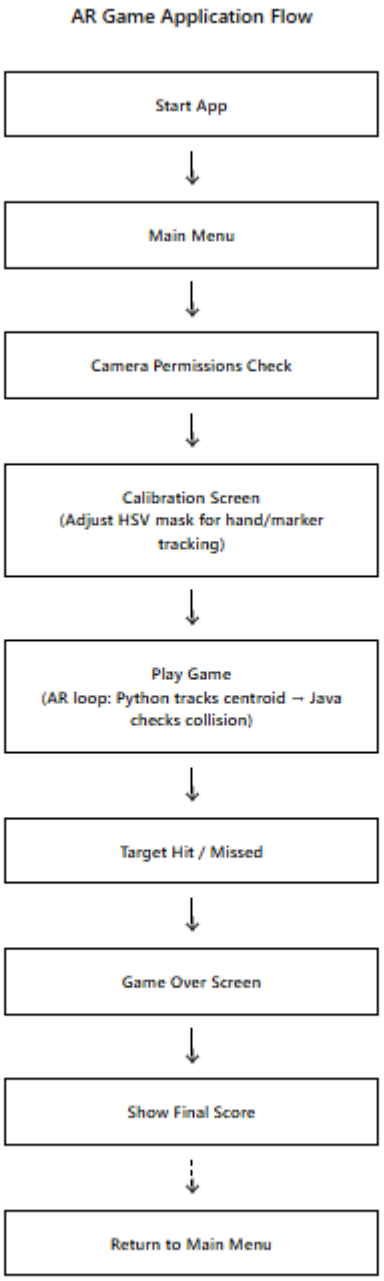

Figure 1. System/Game Flow

XIII. Wireframe / Sketch

Figure 2.1. Debug View

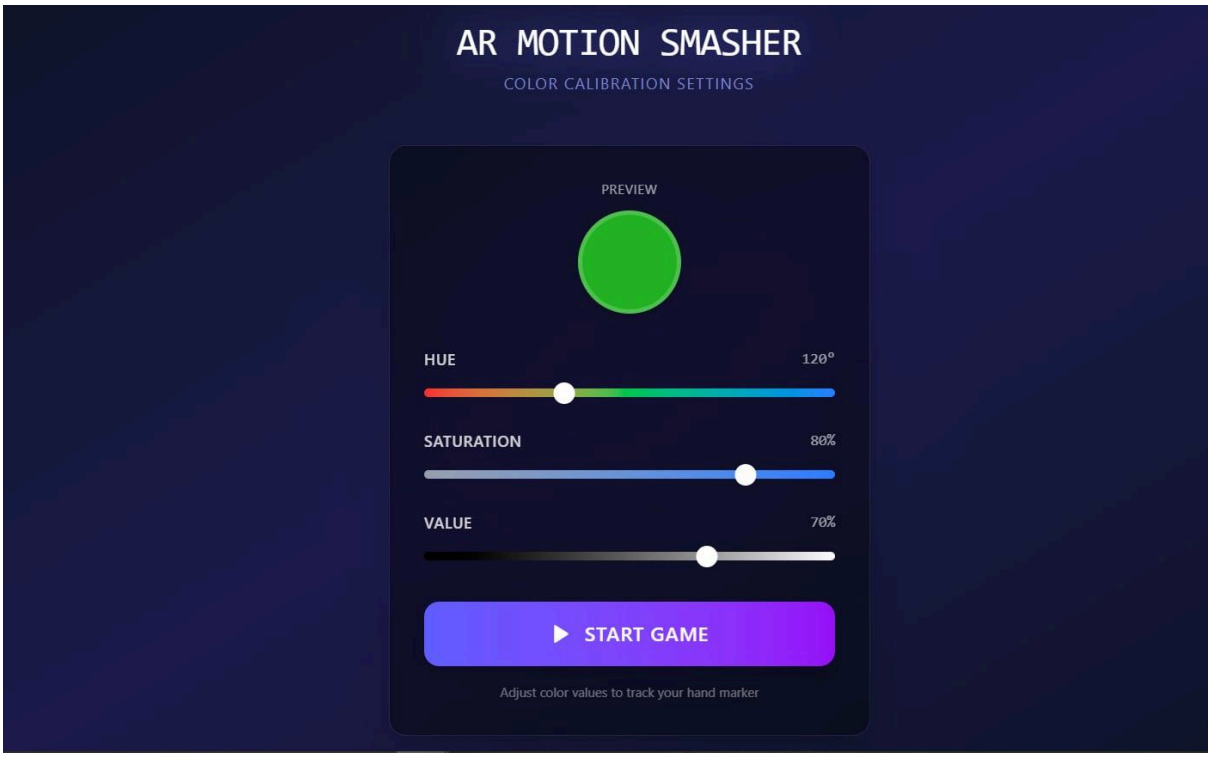

Figure 2.2. Calibration View

Figure 2.3. Game Screen

XIV. Conclusion

AR Motion Smasher's aim is to combat sedentary mobile gaming, which is increasing due to the use of growing numbers of mobile devices. This project will take traditional arcade games and pair them with real-time detection of hand movements to enable users to engage in active and fun ways to play mobile games. It will use the OpenCV processing for complex image processing in Python and process the user's interface with Java, thereby providing a scalable and efficient means of delivering active mobile game play.

XV. References

A. Bin Tomi and D. R. A. Rambli. 2011. A conceptual design for augmented reality games using motion detection as user interface and interaction. In *Visual Informatics: Sustaining Research and Innovations (IVIC 2011)*. Lecture Notes in Computer Science, vol. 7067. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-642-25200-6_29

S. Dhamodaran, et al. 2024. Implementation of hand gesture recognition using OpenCV. *International Journal of Research Publication and Reviews* 5, 5 (2024), 8039–8050.

S. Kim, et al. 2024. ARGo: Augmented reality-based mobile Go stone collision game. *Virtual Reality*. Springer. https://doi.org/10.1007/s10055-023-00919-4

M. C. Parashar, et al. 2020. Real-time hand gesture recognition system for controlling gaming applications using OpenCV and TensorFlow. *International Journal for Multidisciplinary Research (IJFMR)*.

T. A. Vu, et al. 2025. Integrating hand tracking technology for enhanced hand rehabilitation: A Flappy Bird game approach. In *Advances in Data Science and Optimization of Complex Systems (ICAMCS 2024)*. Lecture Notes in Networks and Systems, vol. 1311. Springer, Cham. https://doi.org/10.1007/978-3-031-90606-0_34

M. Wang, et al. 2025. Effectiveness of gamification interventions to improve physical activity and sedentary behavior in children and adolescents: Systematic review and meta-analysis. *JMIR Serious Games* 13 (2025) https://doi.org/10.2196/68151

OpenCV Team. 2024. OpenCV documentation—Image processing in OpenCV. Retrieved https://docs.opencv.org/4.x/d2/d96/tutorial_py_table_of_contents_imgproc.html

Google LLC. 2024. Android CameraX API reference. Android Developers. Retrieved https://developer.android.com/training/camerax

Chaquopy Documentation. 2024. Chaquopy: The Python SDK for Android. Retrieved https://chaquo.com/chaquopy/doc/